

StayKaro AI Caller

Project Status, Architecture & Phased Roadmap

Team handover document based on the current project audit and implementation history

Executive status: The project has a substantial backend and infrastructure foundation, but it is **not yet an end-to-end production AI calling system**. The most important remaining work is connecting the pieces into a real, verified calling flow.

1. What the Product Is

StayKaro AI Caller is intended to be a multi-tenant AI voice-calling platform. The system needs to accept calls through a telephony provider, create and manage a call session, run the voice/AI pipeline, persist call state and results, and expose administration and operational controls.

Target flow: Customer/tenant → telephony provider → voice gateway → internal call API/session layer → speech + AI pipeline → call finalization/storage → admin dashboard and integrations.

2. Overall Project Status

Area	Status	What this means
Infrastructure / containerization	COMPLETED	Core production foundation and routing work is present on main.
Backend API foundation	COMPLETED	Authentication, admin APIs and core API structure are implemented and tested.
Tenant administration	COMPLETED	Tenant CRUD/stats endpoints were added and merged to main.
User response alignment	MERGED	Backend response shape was aligned with frontend expectations and CI was fixed.
Call/session foundation	IMPLEMENTED	SH-03 call-session foundation exists, but it still needs to be connected to a real calling path.
Exotel adapter	IMPLEMENTED / VERIFY	SH-01 has tests passing, but real Exotel sandbox behavior is still unverified.
Real telephony end-to-end	NOT COMPLETE	No verified real call has completed the entire system flow.
STT / TTS / AI voice pipeline	NOT COMPLETE	The complete production voice intelligence path is still outstanding.
Workers / integrations	NOT COMPLETE	Operational background processing and integrations remain.
Admin dashboard	NOT COMPLETE	Frontend/admin product work remains.

Area	Status	What this means
Production readiness	NOT COMPLETE	End-to-end production evidence has not yet been established.

3. Phase Roadmap

Phase 1 — Foundation & Infrastructure

- NH-01 through NH-06 infrastructure/API foundation was completed.
- Production containerization and Nginx/API routing work are present.
- CI includes linting, tests, Docker validation and security checks.
- Status: **COMPLETED**.

Phase 2 — Authentication & Backend Administration

- JWT authentication foundation was implemented.
- Admin APIs cover clients/tenants, calls and users.
- Tenant-facing aliases were added under /admin/tenants with tenant_id string responses.
- User response shapes were aligned with the admin dashboard types.
- Status: **COMPLETED / MERGED**, with known authentication/product gaps still to address.

Phase 3 — Call & Session Foundation

- SH-03 CallSessionManager foundation exists for creating, retrieving, ending and removing call sessions.
- Redis-backed/session-oriented lifecycle work exists.
- Status: **IMPLEMENTED, NOT YET PROVEN END-TO-END**.

Phase 4 — Telephony / Exotel

- SH-01 introduced a provider-neutral telephony abstraction and Exotel provider implementation.
- Webhook authentication, event deduplication and internal-call client code were added.
- A development routing stub exists and is deliberately prevented from becoming a production default.
- 121 tests passed locally on the SH-01 branch after fixes.
- Real Exotel sandbox callback format, credentials, routing and live call behavior still require verification.
- Status: **CODE + TESTS COMPLETE; REAL INTEGRATION VERIFICATION PENDING**.

Phase 5 — Real AI Calling Pipeline

- This is the major unfinished product phase.
- The complete path from live telephony audio through STT, AI reasoning, TTS and back to the caller must be implemented and verified.
- Call lifecycle must be connected to the actual provider and voice gateway.
- Status: **NOT COMPLETE**.

Phase 6 — Workers, Integrations & Operations

- Background workers and integration services still need implementation/integration.
- Production event processing, retries, persistence and operational behavior need verification.

- Status: **NOT COMPLETE.**

Phase 7 — Admin Dashboard

- Frontend-facing API contracts have been considered in backend work.
- The actual admin dashboard experience is not yet considered complete by the audit.
- Status: **NOT COMPLETE.**

Phase 8 — End-to-End & Production Acceptance

- A real call must be initiated, routed, processed by the AI voice pipeline, finalized and visible in the platform.
- Real Redis/PostgreSQL/telephony behavior must be exercised.
- Failure handling, security, retries, observability and deployment behavior must be validated.
- Status: **NOT COMPLETE.**

4. Current Architecture

Component	Role
Nginx	Reverse proxy and API routing foundation.
FastAPI API	Authentication, admin routes and backend services.
PostgreSQL	Persistent application data and call/tenant/user records.
Redis	Session/state infrastructure and supporting runtime state.
Voice Gateway	Telephony/webhook boundary and voice-call orchestration layer.
Provider abstraction	Provider-neutral telephony interfaces under packages/providers/.
Exotel	Current telephony provider adapter under SH-01.
Call Session Manager	Session lifecycle foundation from SH-03.
AI voice pipeline	STT → AI → TTS path; major remaining implementation area.
Workers / integrations	Background processing and external integrations; remaining area.
Admin dashboard	Frontend administration; remaining product area.

5. What Is Actually Verified

- Backend unit/integration test suites have been run successfully for the completed areas.
- Tenant admin API work reached 113 tests before SH-01 additions; the SH-01 branch reached 121 passing tests.
- Ruff checks and formatting checks were clean after the final SH-01 fixes.
- CI Docker, lint and security checks were part of the reviewed workflow.
- The user response alignment PR initially failed CI, was diagnosed, fixed by making Optional Pydantic fields explicitly default to None, and then reached green CI.
- The repository main branch is protected and changes are merged through pull requests.

6. What Is NOT Yet Verified

- A real Exotel sandbox call completing the complete lifecycle.
- Whether the live Exotel callback content/encoding exactly matches the implemented webhook parser.
- Live telephony audio flowing through the gateway into the AI pipeline and back to the caller.
- A production-grade STT/TTS/AI loop.
- Complete worker/integration behavior in a deployed environment.
- Full production deployment and end-to-end acceptance.

7. Important Technical Risks / Open Questions

- Exotel callback payload format must be verified against the real sandbox. The implementation currently treats the callback as JSON, while some Exotel webhook configurations may use form encoding.
- The current Exotel event deduplication trade-off can permanently suppress retries if an event is marked handled before downstream processing succeeds.

- Environment parsing should eventually handle invalid numeric timeout configuration cleanly rather than exposing a raw `ValueError`.
- Authentication and multi-user behavior still require product-level completion and real deployment validation.
- A working API is not proof of a working calling product. The missing evidence is the actual call path.

8. Git / Delivery State

- main is protected. Direct pushes to main are rejected by repository rules.
- Feature work is expected to be pushed to feature branches and merged through pull requests.
- Tenant admin work was merged to main as feat: add tenant admin API endpoints (#14).
- User response alignment was handled through PR #15 and required a CI compatibility fix before merge.
- SH-01 Exotel work was developed on the sh-01-exotel-adapter branch and pushed for PR review.
- Before starting new work, always synchronize local main with origin/main.

9. Recommended Engineering Order From Here

#	Next step	Reason
1	Finish / verify the internal call lifecycle contract	The voice gateway needs a reliable internal API for call creation, retrieval and finalization.
2	Verify Exotel in a real sandbox	Confirm credentials, webhook format, token behavior, call routing and callback lifecycle.
3	Connect real telephony to the session layer	Prove connected → session created → media/voice processing → completed → cleanup.
4	Implement the AI voice pipeline	Integrate STT, AI reasoning and TTS with the live call path.
5	Implement workers/integrations	Add durable background processing, retries and external-system integration.
6	Build the admin dashboard	Consume the already-defined backend contracts and expose operational controls.
7	Run full end-to-end acceptance	Use real infrastructure and a real call to prove the complete product.
8	Production hardening	Observability, security, failure handling, scaling and deployment verification.

10. Definition of 'Done'

For this project, a ticket should not be treated as product-complete merely because code exists or local tests pass. A feature is complete when its implementation, automated tests, integration behavior and acceptance criteria are satisfied, and when the required real-environment behavior has been demonstrated.

- Code implemented.
- Automated tests passing.
- CI checks passing.
- Integration contract verified.

- Real external dependency verified where applicable.
- End-to-end behavior demonstrated where applicable.
- Security and failure behavior reviewed.
- Merged into the protected main branch.

11. Bottom Line for the New Team

Do not describe the project as finished. The correct description is: **the platform foundation and significant backend functionality are implemented, while the core real-world AI calling path remains to be completed and proven.**

The highest-value milestone now is a single reliable real call that traverses the complete system. Once that works, the team can expand reliability, scale, dashboard functionality and integrations around a proven core.

Source basis: current project audit/report and implementation history supplied for this handover. Status labels intentionally distinguish implementation from real-world verification.